

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3?

129

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

Because interface S0/0/1 on R2 is configured as passive, R3 cannot form a neighbor adjacency directly with R2. However, R3 still learns the route to 192.168.2.0/24 via R1, which has an active neighbor relationship with R2.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

65 (SSerial link cost 64 + Ethernet cost 1)

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

__Because interface S0/0/1 on R2 is configured as passive, R3 cannot form a neighbor adjacency directly with R2. However, R3 still learns the route to 192.168.2.0/24 via R1, which has an active neighbor relationship with R2.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

__For 192.168.3.0/24: R1's S0/0/1 interface is configured with a bandwidth of 128 Kbps, and the corresponding OSPF cost is 781. R3's G0/0 interface has a default cost of 1. So, the total accumulated cost is $781 + 1 = 782$.

__For 192.168.23.0/30: R1's S0/0/0 and S0/0/1 are both configured with a bandwidth of 128 Kbps, costing 781. The serial interfaces of R2 and R3 connected to 192.168.23.0/30 retain the default bandwidth of 1544 Kbps, costing 64. The total cost of the two equal-cost paths to 192.168.23.0/30 is $781 + 64 = 845$.

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

__All serial interfaces in the topology are configured with a bandwidth of 128 Kbps, and the corresponding OSPF cost of each serial interface is 781. The 192.168.23.0/30 network is a point-to-point link between R2 and R3, and R1 needs to pass through two serial links to reach this network. The accumulated cost is $781 + 781 = 1562$.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

__ OSPF selects the path with the lowest accumulated cost as the optimal route. We manually configured ip ospf cost 1565 on R1's S0/0/1 interface, so the cost of the direct path R1→R3 to 192.168.3.0/24 is $1565 + 1(R3 G0/0) = 1566$. The cost of the indirect path R1→R2→R3 is $781 + 781 + 1 = 1563$, which is lower than the direct path. Therefore, OSPF selects the indirect path through R2 as the optimal route to 192.168.3.0/24.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

__ The router ID uniquely identifies a router in the OSPF domain. If not manually set, it is automatically chosen based on the highest loopback or physical IP address, which can lead to unpredictable changes if interfaces go down. Manual assignment ensures stability, consistency, and easier troubleshooting.

2. Why is the DR/BDR election process not a concern in this lab?

__ All OSPF links in this topology are point-to-point serial connections. DR/BDR election only occurs on multi-access networks where multiple routers share a common segment. Point-to-point links do not require a DR/BDR.

3. Why would you want to set an OSPF interface to passive?

__ Setting an interface to passive prevents OSPF from sending hello messages on that interface, which reduces unnecessary traffic, improves security, and is typically done on LAN interfaces that connect to end devices where no OSPF neighbor exists.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

__ The use of areas, particularly area 0, allows OSPF to summarize routes and limit the propagation of Link State Advertisements. This hierarchical design reduces routing table sizes and processing overhead, enabling scalability.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

__ network 10.0.0.0 0.255.255.255 area 0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

__ The route from EIGRP will be used because it has the lowest administrative distance. EIGRP AD is 90, OSPF AD is 110, and RIP AD is 120. The routing decision is based on AD first, not the metric.